Comparison of fatty acid, cholesterol, and vitamin A and E composition in eggs from hens housed in conventional cage and range production facilities.

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The public perceives that the nutritional quality of eggs produced as free range is superior to that of eggs produced in cages. Therefore, this study compared the nutrient content of free-range vs. cage-produced shell eggs by examining the effects of the laboratory, production environment, and hen age. A flock of 500 Hy-Line Brown layers were hatched simultaneously and received the same care (i.e., vaccination, lighting, and feeding regimen), with the only difference being access to the range. The nutrient content of the eggs was analyzed for cholesterol, n-3 fatty acids, saturated fat, monounsaturated fat, polyunsaturated fat, β-carotene, vitamin A, and vitamin E. The same egg pool was divided and sent to 4 different laboratories for analysis. The laboratory was found to have a significant effect on the content of all nutrients in the analysis except for cholesterol. Total fat content in the samples varied ($P < 0.001$) from a high of 8.88% to a low of 6.76% in laboratories D and C, respectively. Eggs from the range production environment had more total fat ($P < 0.05$), monounsaturated fat ($P < 0.05$), and polyunsaturated fat ($P < 0.001$) than eggs produced by caged hens. Levels of n-3 fatty acids were also higher ($P < 0.05$), at 0.17% in range eggs vs. 0.14% in cage eggs. The range environment had no effect on cholesterol (163.42 and 165.38 mg/50 g in eggs from caged and range hens, respectively). Vitamin A and E levels were not affected by the husbandry to which the hens were exposed but were lowest at 62 wk of age. The age of the hens did not influence the fat levels in the egg, but cholesterol levels were highest ($P < 0.001$) at 62 wk of age (172.54 mg/50 g). Although range production did not influence the cholesterol level in the egg, there was an increase in fat levels in eggs produced on the range.